# Dog bite injuries and armed conflict-related environmental stressors: a nationwide population-based time-series study

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# Abstract

**Background**: Dog bite injuries pose a substantial public health burden worldwide. Environmental and acoustic stressors may contribute to canine behavioral dysregulation. However, there are as yet no population-level studies evaluating whether the incidence of dog bites increases during periods of armed conflict. The present study was conducted in Israel where the population is routinely exposed to an episodic pattern of high-intensity conflicts (escalation) alternating with periods of calm (de-escalation), providing a unique natural experiment to examine the effect of environmental stressors on population-level modifiers of injury risk.

**Methods**: This nationwide retrospective observational study covered the decade from 2014 to 2025. Healthcare-encounter data were used to capture dog bite-related diagnoses in both hospital and community settings. Exposure to armed conflict-related environmental stress was operationalized using the number of civil defense sirens per month, categorized as none (0), low (<500), or high (≥500) and aggregated by geographic region (North, Center, South). Monthly dog bite counts were modeled using negative binomial regression adjusted for region, seasonality, age group, sex, and socioeconomic status. Secondary outcomes were hospitalization within 7 days and surgical intervention within 30 days, reflecting injury severity.

**Results**: A total of 63,285 dog bite-related encounters were identified. Compared to months with no sirens, adjusted dog bite incidence increased by 15% during low-exposure months (IRR 1.15, 95% CI 1.13-1.18) and by 33% during high-exposure months (IRR 1.33, 95% CI 1.28-1.37), demonstrating a graded exposure-response association.

High exposure was associated with increased odds of surgical intervention within 30 days (OR 1.09, 95% CI 1.02-1.16; *P*=0.013).

**Conclusions**: This study provides the first population-level evidence linking armed conflict-related environmental stressors to increased dog bite incidence, using a quantitative graded exposure measure rather than a binary conflict-period definition. Dog bite prevention and healthcare preparedness should be taken into consideration in civilian injury mitigation strategies during armed conflict.

# Background

Dog bites are a significant and growing public health concern worldwide, with millions of cases reported annually. In the United States, dog bites rank among the leading causes of nonfatal emergency department visits, with approximately 344,000 encounters per year and an incidence of up to 225 per 100,000 population [1]. Dog bite injuries pose a substantial physical, psychological, and economic burden, particularly among pediatric populations [2]. The etiology is multifactorial, involving dog-specific characteristics such as breed, sex, training, and behavioral traits, in addition to human behavior and environmental stressors that may alter canine responses. The stress exposure may be either acute or chronic [3, 4].

Among the potential environmental stressors, several studies have identified loud noises as major contributing factors to anxiety-related and behavior dysregulation and aggression in dogs [5, 6, 7]. The most common noises that elicit fear responses are high-decibel fireworks, thunderstorms, sirens, and explosions [7, 8]. Experimental studies examining controlled acoustic environments reported physiological stress responses in dogs, including elevated cortisol levels and increased heart rate, supporting the biological plausibility of noise-induced behavioral changes [8, 9, 10].

Besides such external stimuli, dogs are sensitive to their living conditions and the emotional states and behavioral patterns of their owners. Heightened stress, fear, and anxiety among caregivers may contribute to altered animal handling, reduced tolerance, and more unpredictable interactions, potentially lowering the threshold for aggressive canine responses. During the COVID-19 pandemic, multiple studies reported an increase in dog bite injuries, particularly among children, which were attributed to disrupted routines, increased household stress, reduced supervision, and prolonged close contact between people and pets [11, 12]. These observations underscore how sustained environmental and psychosocial stressors can influence human-animal interactions and increase the risk of dog bite injuries.

Beyond identifying individual-level determinants, dog-bite injuries can be studied using an ecological injury-risk model, examining changes in exposure patterns and susceptibility as a function of multiple, interacting factors across different levels of organization within whole environmental systems. In this context, armed conflict represents a unique population-level stressor capable of modifying the risk of dog bite injury through different, concurrent, pathways. It may simultaneously alter physical environmental conditions (e.g., high-intensity auditory stimuli including civil defense sirens, explosions, gunfire, and noise generated by aircraft, drones, and incoming or outgoing projectiles), human behavior and routines (e.g., time spent at home and supervision patterns), and psychosocial vulnerability (e.g., heightened anxiety and emotional tension). The stressors may occur as discrete acute events or as repeated and prolonged episodes over weeks or months. Accordingly, research conducted in disaster areas revealed significant behavioral changes in companion animals including increased anxiety, heightened reactivity, and unpredictable conduct [13–15].

Taken together, these findings suggest that armed conflict-related environmental stressors may increase both human and canine stress, thereby modifying human-animal interaction dynamics and increasing the likelihood of bite events, potentially with greater injury severity.

Despite the growing recognition of the behavioral effects of environmental stressors on companion animals, empirical data on the relationship between armed conflict-related environmental stress and dog bite incidence remain limited. To our knowledge, there are no published systematic, population-level studies of armed conflict stressors as modifiers of injury risk. Israel's recurrent exposure to fluctuating intensities of armed conflict over the past decade, characterized by intermittent periods of escalation interspersed with intervals of relative calm, provides a unique natural experiment to examine this issue. Moreover, the nationwide availability of civil defense siren records makes it possible to use air raid alerts as a precise, objective proxy to quantify acute and cumulative environmental stress exposure, overall and across time and geography.

The primary aim of the present study was to evaluate the association between armed conflict-related environmental stress and the incidence of dog bite injuries requiring medical evaluation using civil defense siren activity as an objective proxy for periods of escalation. The secondary aim were to determine whether the level of stressor is related to the severity of injury, as reflected by the need for hospital admission or operative management.

## Methods

A retrospective, nationwide, observational study was conducted, spanning the decade from 2014 to 2025, during which the Israeli population was exposed to a cycle of high-intensity military conflicts and heightened civil defense siren activity (escalation) interspersed with intervals of calm (de-escalation). Healthcare encounter data were extracted at the end of the study from the research data-sharing platform (powered by MDClone) of Clalit Health Services (CHS), Israel's largest healthcare organization. CHS insures nearly half the national population, with broad geographic coverage. Its electronic health records databases include longitudinal, de-identified information from multiple care sources. For the present study, dog bite cases were identified by ICD-9 diagnostic codes for dog bite. Case ascertainment was not restricted to emergency department encounters; dog bite diagnoses recorded in community-based settings, including primary care, urgent care, and outpatient clinic visits, were captured as well. All individuals of any age with a documented dog bite diagnosis during the study period were included. Given the use of anonymized, population-level administrative data, no exclusion criteria were applied.

Data on civil defense siren activity were obtained from the Israel Home Front Command. Siren events were recorded by date and geographic region (North, Center, South) and aggregated on a monthly basis for each region. Siren activity was categorized into three predefined groups: months with no sirens, months with < 500 sirens (low exposure), and months with ≥ 500 sirens (high exposure). At the national level, the incidence of dog bite diagnoses was compared across these categories to assess whether months characterized by exposure to more civil defense sirens were associated with higher rates of dog bite injuries. Subsequently, the same analyses were stratified by geographic region (North, Center, South) to evaluate whether regional increases in exposure to civil defense sirens were associated with a higher regional dog bite incidence during the corresponding months.

Injury severity was assessed using two clinically meaningful endpoints: hospital admission within 7 days of the index dog bite encounter and surgical intervention within 30 days. Relevant surgical interventions were defined as procedures requiring management in the operating room, including wound debridement, operative wound closure, skin grafting, and local, regional, or free-flap reconstruction. Surgical procedures performed within 30 days of the initial dog bite diagnosis were assumed to be related to the index injury. The need for operative intervention and/or hospital admission was used as a proxy for more severe injuries, reflecting greater tissue damage and clinical complexity.

The primary outcome measure of the study was the monthly incidence of dog bite-related healthcare encounters, assessed at both the national and regional levels and evaluated in relation to monthly exposure to siren activity. The secondary outcome measure was injury severity in relation to level of siren exposure.

Monthly counts of dog bite-related healthcare encounters were modeled using negative binomial regression to account for overdispersion in the count data (variance exceeding the mean), for which Poisson assumptions were not met. The dependent variable was the monthly number of dog bite incidents. The primary exposure was monthly siren activity, categorized a priori by level of exposure (0, < 500, or ≥ 500 sirens). Models included geographic region and were adjusted for seasonality and relevant demographic and socioeconomic covariates, including age group, sex, and socioeconomic status. Incidence rate ratios (IRRs) with 95% confidence intervals (CIs) were reported. Statistical analyses were performed using R (version 4.3.2), and statistical significance was defined as a two-sided $\alpha$ level of 0.05.

This study was based on the analysis of anonymized, retrospective administrative healthcare data and did not involve direct contact with human or animal subjects. The study was approved by the Institutional Review Board (Helsinki Committee) of Rabin Medical Center.

## Results

A total of 63,285 dog bite-related healthcare encounters were identified during the study period, of which 36,661 occurred during months with no sirens, 18,971 during months with low siren exposure (< 500 sirens), and 7,653 during months with high siren exposure (≥ 500 sirens). The demographic, socioeconomic, geographic, seasonal, and clinical characteristics of the dog bite cases stratified by siren exposure status are summarized in Table 1.

Table 1
Patient, environmental, and dog-bite characteristics by level of siren exposure*.

| Variable | Overall n = 63,285 | No sirens n = 36,661 | Low exposure n = 18,971 | High exposure n = 7,653 |
|---|---|---|---|---|
| Age group (years) | | | | |
| 0–4 | 4,825 (7.6%) | 2,667 (7.3%) | 1,571 (8.3%) | 587 (7.7%) |
| 5–14 | 16,572 (26%) | 9,523 (26%) | 5,184 (27%) | 1,865 (24%) |
| 15–24 | 10,101 (16%) | 5,919 (16%) | 2,989 (16%) | 1,193 (16%) |
| 25–44 | 15,871 (25%) | 9,290 (25%) | 4,637 (24%) | 1,944 (25%) |
| 45–64 | 9,805 (15%) | 5,716 (16%) | 2,870 (15%) | 1,219 (16%) |
| >65 | 6,111 (9.7%) | 3,546 (9.7%) | 1,720 (9.1%) | 845 (11%) |
| Sex | | | | |
| Female | 25,164 (40%) | 14,357 (39%) | 7,573 (40%) | 3,234 (42%) |
| Male | 38,121 (60%) | 22,304 (61%) | 11,398 (60%) | 4,419 (58%) |
| Socioeconomic status† | | | | |
| Low | 12,325 (19%) | 7,236 (20%) | 3,773 (20%) | 1,316 (17%) |
| Medium | 34,131 (54%) | 19,812 (54%) | 10,082 (53%) | 4,237 (55%) |
| High | 12,569 (20%) | 7,510 (20%) | 3,453 (18%) | 1,606 (21%) |
| No data | 4,260 (6.7%) | 2,103 (5.7%) | 1,663 (8.8%) | 494 (6.5%) |
| Region | | | | |
| South | 9,141 (14%) | 1,641 (4.5%) | 6,623 (35%) | 877 (11%) |
| Center | 33,150 | 20,375 | 8,124 (43%) | 4,651 (61%) |

Notes: values are presented as n (%).

* Siren exposure was defined by the number of monthly regional civil defense sirens: none (0), low (< 500), and high (≥ 500).

†Socioeconomic status was classified according to the national socioeconomic index.

| Variable | Overall<br>n = 63,285 | No sirens<br>n = 36,661 | Low exposure<br>n = 18,971 | High exposure<br>n = 7,653 |
|---|---|---|---|---|
| | (52%) | (56%) | | |
| North | 20,994 (33%) | 14,645 (40%) | 4,224 (22%) | 2,125 (28%) |
| Season of injury | | | | |
| Spring | 17,012 (27%) | 9,625 (26%) | 5,362 (28%) | 2,025 (26%) |
| Summer | 17,048 (27%) | 9,632 (26%) | 5,241 (28%) | 2,175 (28%) |
| Autumn | 15,633 (25%) | 8,625 (24%) | 4,245 (22%) | 2,763 (36%) |
| Winter | 13,592 (21%) | 8,779 (24%) | 4,123 (22%) | 690 (9.0%) |
| Hospitalization within 7 days | 452 (0.7%) | 262 (0.7%) | 131 (0.7%) | 59 (0.8%) |
| Surgical intervention within 30 days | 11,389 (18%) | 6,591 (18%) | 3,382 (18%) | 1,416 (19%) |
| Notes: values are presented as n (%). | | | | |
| * Siren exposure was defined by the number of monthly regional civil defense sirens: none (0), low (< 500), and high (≥ 500). | | | | |
| †Socioeconomic status was classified according to the national socioeconomic index. | | | | |

Adjusted negative binomial regression models revealed an increase in dog bite incidence both in months with low siren exposure (IRR 1.15, 95% CI 1.13–1.18; $P < 0.001$) and in months with high siren exposure (IRR 1.33, 95% CI 1.28–1.37; $P < 0.001$) compared to months with no sirens (Table 2). The temporal patterns and exposure categorization are shown in Fig. 1. The regional stratification by exposure category is presented in Fig. 2.

Table 2
Association between siren exposure level and monthly dog bite incidence

| Variable | IRR | CI 95% | *P*-value |
|---|---|---|---|
| Siren exposure category | | | |
| No siren | Reference | - | - |
| Low exposure (< 500 sirens) | 1.15 | 1.13–1.18 | < 0.001 |
| High exposure (≥ 500 sirens) | 1.33 | 1.28–1.37 | < 0.001 |
| Notes: IRRs estimated using negative binomial regression. | | | |

**Figure 1.**

**Figure 2.**

Regarding secondary outcomes, 452 (0.7%) encounters were followed by hospitalization within 7 days and 11,389 (18.0%) by surgical intervention within 30 days (Table 1). In adjusted models, high siren exposure was associated with increased odds of surgical intervention within 30 days (OR 1.09, 95% CI 1.02-1.16; *P*=0.013). No statistically significant associations were observed for hospitalization within 7 days (Table 3).

Table 3
Association between siren exposure level and injury severity among dog bite cases

| Outcome | Siren exposure category | Adjusted OR | CI 95% | *P*-value |
|---|---|---|---|---|
| Surgical intervention within 30 days | No siren | Reference | - | - |
| | Low exposure (< 500 sirens) | 1.04 | 0.99–1.10 | 0.11 |
| | High exposure (≥ 500 sirens) | 1.09 | 1.02–1.16 | 0.013 |
| Hospitalization within 7 days | No siren | Reference | - | - |
| | Low exposure (< 500 sirens) | 1.06 | 0.85–1.33 | 0.60 |
| | High exposure (≥ 500 sirens) | 1.09 | 0.81–1.44 | 0.60 |
| Notes: Odds ratios estimated using multivariable logistic regression models. | | | | |
| Abbreviations: OR, odds ratio; CI, confidence interval. | | | | |

## Discussion

In this nationwide retrospective study (2014-2025), we evaluated whether armed conflict-related environmental stressors, operationalized using civil defense siren activity, were associated with population-level variations in dog bite-related healthcare encounters. We identified a statistically significant graded association between the level of siren exposure and monthly dog bite incidence. In adjusted negative binomial models, compared to months with no sirens, months with low siren exposure (<500 sirens) were associated with a 15% increase in monthly dog bite incidence (IRR 1.15, 95% CI 1.13-1.18), and months with high exposure (≥500 sirens), with a 33% increase (IRR 1.33, 95% CI 1.28-1.37). Importantly, this study provides, to our knowledge, the first national-level evidence linking armed conflict-related environmental stressors to increased dog bite incidence.

A key strength of this work is the use of siren alert data as a quantitative exposure measure. Civil defense sirens provide an objective, time-stamped, region-specific, and publicly documented indicator of environmental stress related to conflict escalation. Siren activity captures not only acute acoustic exposure but also broader contextual disruption during escalation periods, including shelter confinement, altered routines, and heightened mental strain, allowing for the assessment of population-level injury risk within a natural experiment framework. Notably, the graded siren exposure categorization offers a reproducible approach that extends beyond binary definitions of conflict versus non-conflict periods and enables evaluation of dose-response patterns at the population level.

The observed association is biologically and behaviorally plausible [8-10]. High-intensity auditory stimuli have been associated with anxiety-related behavioral dysregulation in dogs, and companion animals are also known to be sensitive to changes in household stress and routine [5-7]. However, the present study was not designed to isolate a single mechanistic pathway. Rather, it adopts an ecological injury-risk perspective in which escalation-related stressors are expected to act simultaneously across multiple domains, including environmental context, human behavior and supervision patterns, and canine stress reactivity, resulting in a net population-level change in dog bite risk. Importantly, these pathways are not mutually exclusive and likely co-occur during periods of heightened threat exposure.

Additionally, we assessed clinically meaningful severity endpoints reflecting healthcare utilization and system burden rather than behavior alone. Hospitalization within 7 days was uncommon and not significantly associated with siren exposure category. However, high-exposure months were associated with increased odds of surgical intervention within 30 days (OR 1.09, 95% CI 1.02-1.16). Although the effect size was small, it may be operationally relevant given the high baseline volume of dog bite encounters. Even modest increases in operative management may translate into a measurable surgical workload during periods when healthcare systems are simultaneously managing many escalation-related demands.

Regional differences were observed in the adjusted models, with lower dog bite incidence in the North and South of the country compared with the Central region. The descriptive patterns shown in Figure 2 similarly indicate higher monthly counts of dog bite-related encounters in the Central region over the study period. However, the interpretation of geographic differences in dog bite incidence in Israel requires careful consideration of population distribution and displacement dynamics during escalation periods. The Central region includes a substantial proportion of the national population and has a higher population density, which may influence baseline exposure patterns and healthcare utilization. In addition, in Israel, escalation periods have historically involved a disproportionate siren burden in the South and, more recently, in the North. They may also involve substantial internal displacement and temporary relocation of residents from conflict-adjacent areas to central regions, altering both the geographic denominators at risk and the location of healthcare utilization. Thus, dog bite encounters may be redistributed toward the Central region irrespective of where exposure occurred. Consequently, region-level comparisons of dog bite incidence should be interpreted as reflecting both underlying demographic structure and time-varying population movement rather than stable geographic differences in risk.

These findings have actionable public health and injury prevention implications. Preparedness during escalation periods, particularly in high-exposure areas, may benefit from incorporating dog bite injury mitigation into emergency department and surgical resource planning. Public guidance issued during siren events could include practical dog-handling recommendations, emphasizing child supervision, minimization of high-risk interactions in confined environments such as shelters, and use of restraints when feasible. Dog bite prevention may therefore represent a previously underrecognized component of civilian injury prevention during armed conflict.

Several limitations merit consideration. This study relied on diagnostic coding and captured dog bite events resulting in healthcare utilization. Under-ascertainment of mild injuries is likely in routine periods and may be more pronounced during escalation periods when individuals may avoid seeking care for minor wounds. Such misclassification would be expected to be largely nondifferential and to bias effect estimates toward the null, suggesting that the observed associations may be conservative. In addition, data on dog breed, ownership status, household structure, and individual-level behavioral factors were unavailable, limiting mechanistic inference. Finally, the ecological design does not permit causal attribution at the individual level, and residual confounding may persist despite adjustment for demographics, seasonality, and region.

In summary, using a nationwide dataset spanning more than a decade, we demonstrate a graded association between escalation-related environmental stress, quantified through civil defense siren burden, and population-level dog bite incidence. By leveraging a quantitative exposure measure rather than a binary conflict definition, this study strengthens evidence that armed conflict-related environmental stressors can modify civilian injury risk. These findings support integrating dog bite prevention and healthcare preparedness into civilian injury mitigation strategies during armed conflict.

# Abbreviations

| | |
|---|---|
| CHS | Clalit Health Services |

# Declarations

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Not applicable.

**Author contributions**

S.S. conceived and designed the study, led data collection, performed the data analysis, interpreted the findings, and drafted the manuscript.

L.S. contributed to data collection and data analysis.

A.T. and S.F. provided statistical analysis and methodological support.

D.A.-E. supervised the study, contributed to the interpretation of the results, and critically reviewed the manuscript for important intellectual content.

A.O. supervised the study, contributed to the study design and interpretation of the results, and took a leading and full role in drafting and revising the manuscript.

All authors reviewed and approved the final manuscript.

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**Data availability**

The datasets analyzed during the current study are based on anonymized administrative healthcare data and are subject to institutional data access regulations. Therefore, the data are not publicly available but may be accessed upon reasonable request and with appropriate institutional approvals.

**Ethics approval and consent to participate**

The study was conducted in adherence of the Declaration of Helsinki and approved by the Institutional Review Board of Rabin Medical Center (approval #RMC-0476-25) which waived the need for patient consent owing to the retrospective study design.

**Consent for publication**

Not applicable.

**Competing interests**

The authors declare no competing interests.

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# Figures

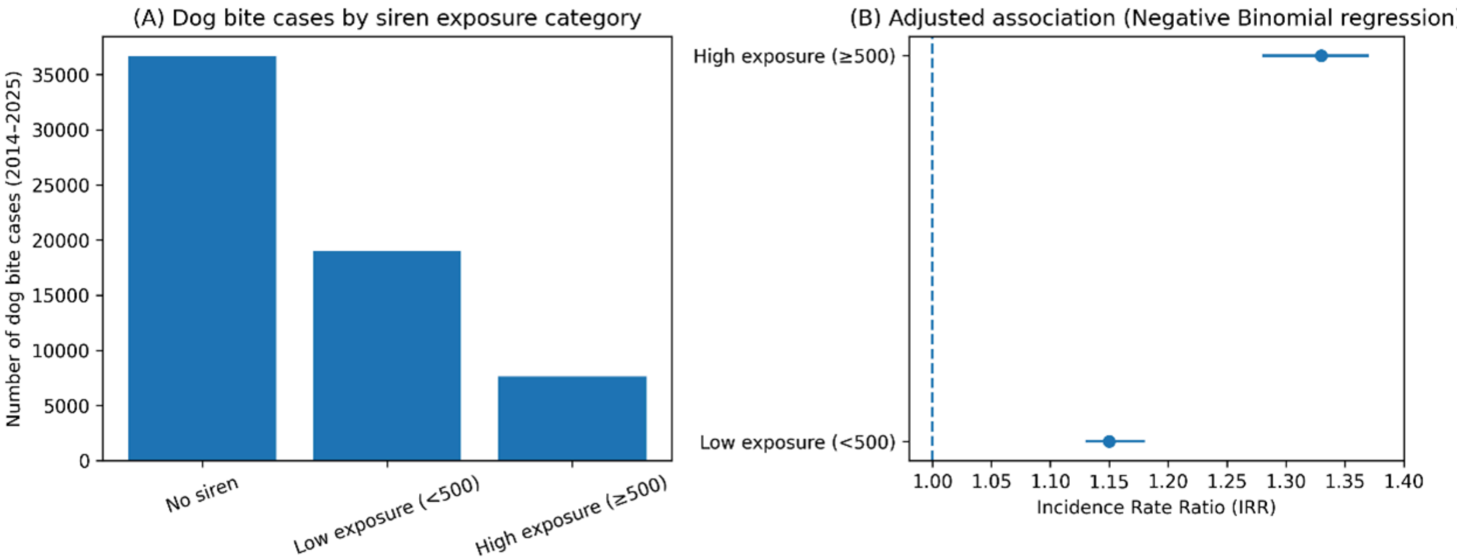


**Figure 1**

Dog bite-related healthcare encounters by siren exposure category in Israel (2014-2025). (A) Total number of dog bite-related healthcare encounters aggregated across the study period, stratified by monthly civil defense siren exposure: no sirens, low exposure (<500 sirens per month), and high exposure (≥500 sirens per month). (B) Adjusted incidence rate ratios (IRRs) with 95% confidence intervals (CIs) for dog bite incidence by siren exposure category, estimated using multivariable negative binomial regression models. Months with no sirens served as the reference category (IRR = 1). Models were adjusted for geographic region, seasonality, age group, sex, and socioeconomic status.


**Figure 2**

Monthly dog bite-related healthcare encounters by geographic region and siren exposure category in Israel (2014-2025).

Bars represent monthly counts of dog bite-related healthcare encounters stratified by geographic region (Center, North, South) and level of civil defense siren exposure: no sirens, low exposure (<500 sirens per month), and high exposure (≥500 sirens per month). This figure illustrates temporal and regional variation in dog bite incidence in relation to escalation-related environmental stress.